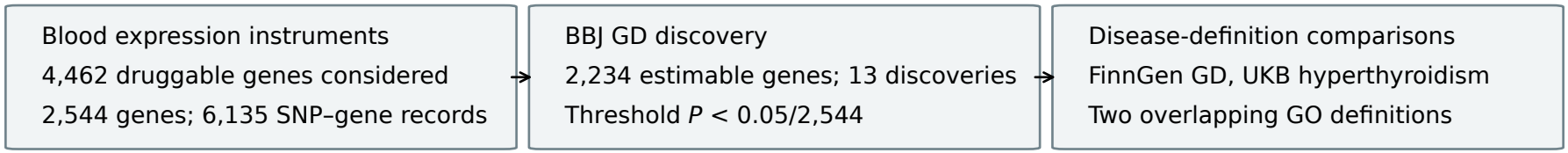
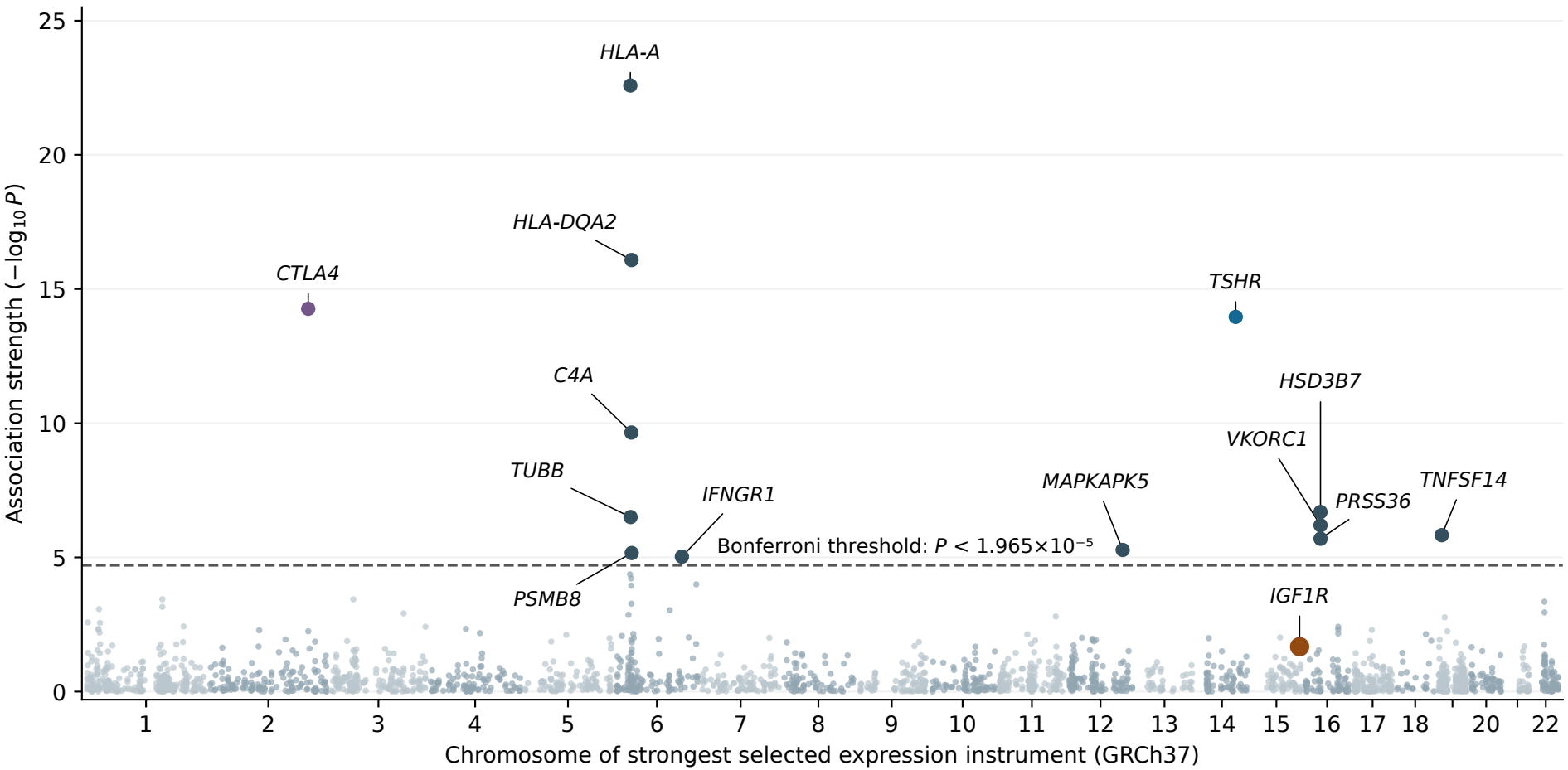
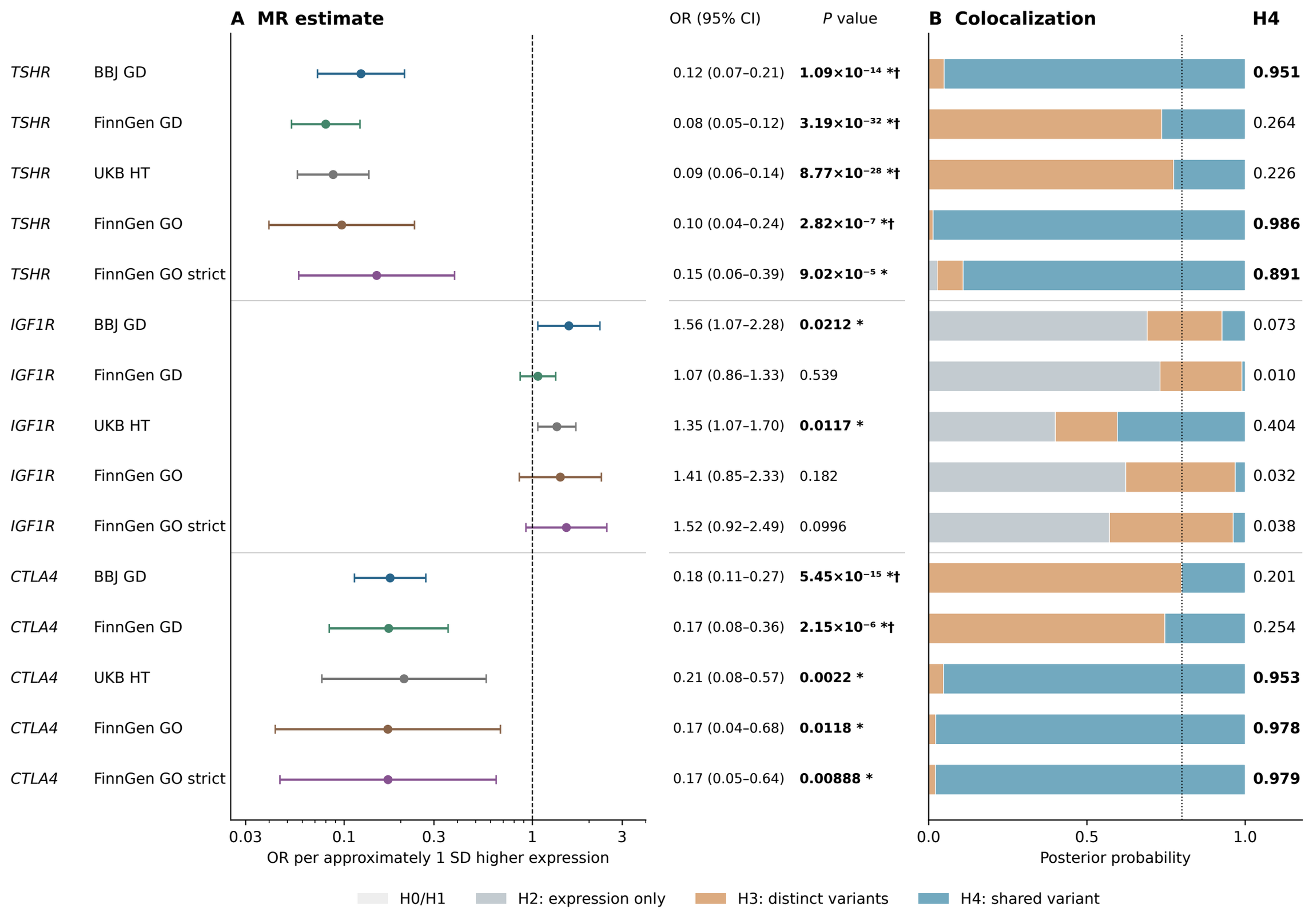
A Study design



Regional follow-up: *TSHR* and *IGF1R* at the center; *CTLA4* comparator; 24 selected genes in total.

B BBJ gene-level MR associations

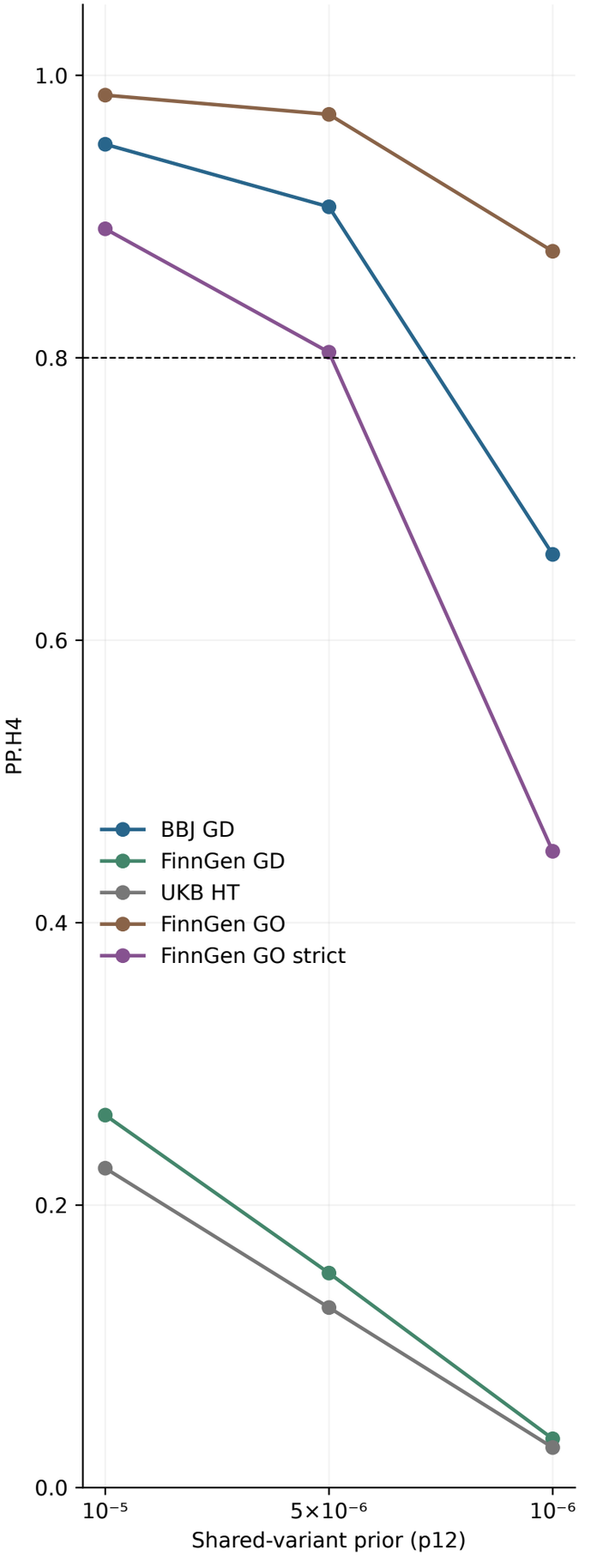


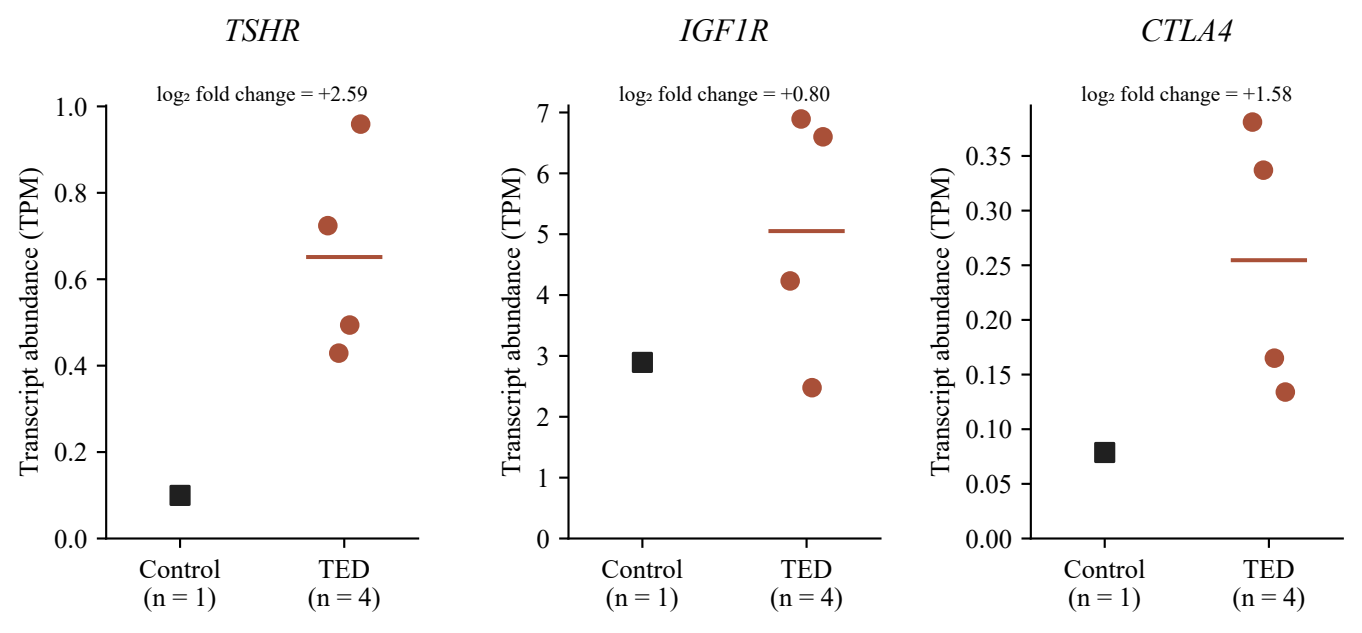

A MR estimate
OR (95% CI)
P value
B Colocalization
H4
TSHR BBJ GD
0.12 (0.07–0.21)
1.09×10⁻¹⁴ *†
0.951
TSHR FinnGen GD
0.08 (0.05–0.12)
3.19×10⁻³² *†
0.264
TSHR UKB HT
0.09 (0.06–0.14)
8.77×10⁻²⁸ *†
0.226
TSHR FinnGen GO
0.10 (0.04–0.24)
2.82×10⁻⁷ *†
0.986
TSHR FinnGen GO strict
0.15 (0.06–0.39)
9.02×10⁻⁵ *
0.891
IGF1R BBJ GD
1.56 (1.07–2.28)
0.0212 *
0.073
IGF1R FinnGen GD
1.07 (0.86–1.33)
0.539
0.010
IGF1R UKB HT
1.35 (1.07–1.70)
0.0117 *
0.404
IGF1R FinnGen GO
1.41 (0.85–2.33)
0.182
0.032
IGF1R FinnGen GO strict
1.52 (0.92–2.49)
0.0996
0.038
CTLA4 BBJ GD
0.18 (0.11–0.27)
5.45×10⁻¹⁵ *†
0.201
CTLA4 FinnGen GD
0.17 (0.08–0.36)
2.15×10⁻⁶ *†
0.254
CTLA4 UKB HT
0.21 (0.08–0.57)
0.0022 *
0.953
CTLA4 FinnGen GO
0.17 (0.04–0.68)
0.0118 *
0.978
CTLA4 FinnGen GO strict
0.17 (0.05–0.64)
0.00888 *
0.979
0.03
0.1
0.3
1
3
OR per approximately 1 SD higher expression
0.0
0.5
1.0
Posterior probability
H0/H1
H2: expression only
H3: distinct variants
H4: shared variant

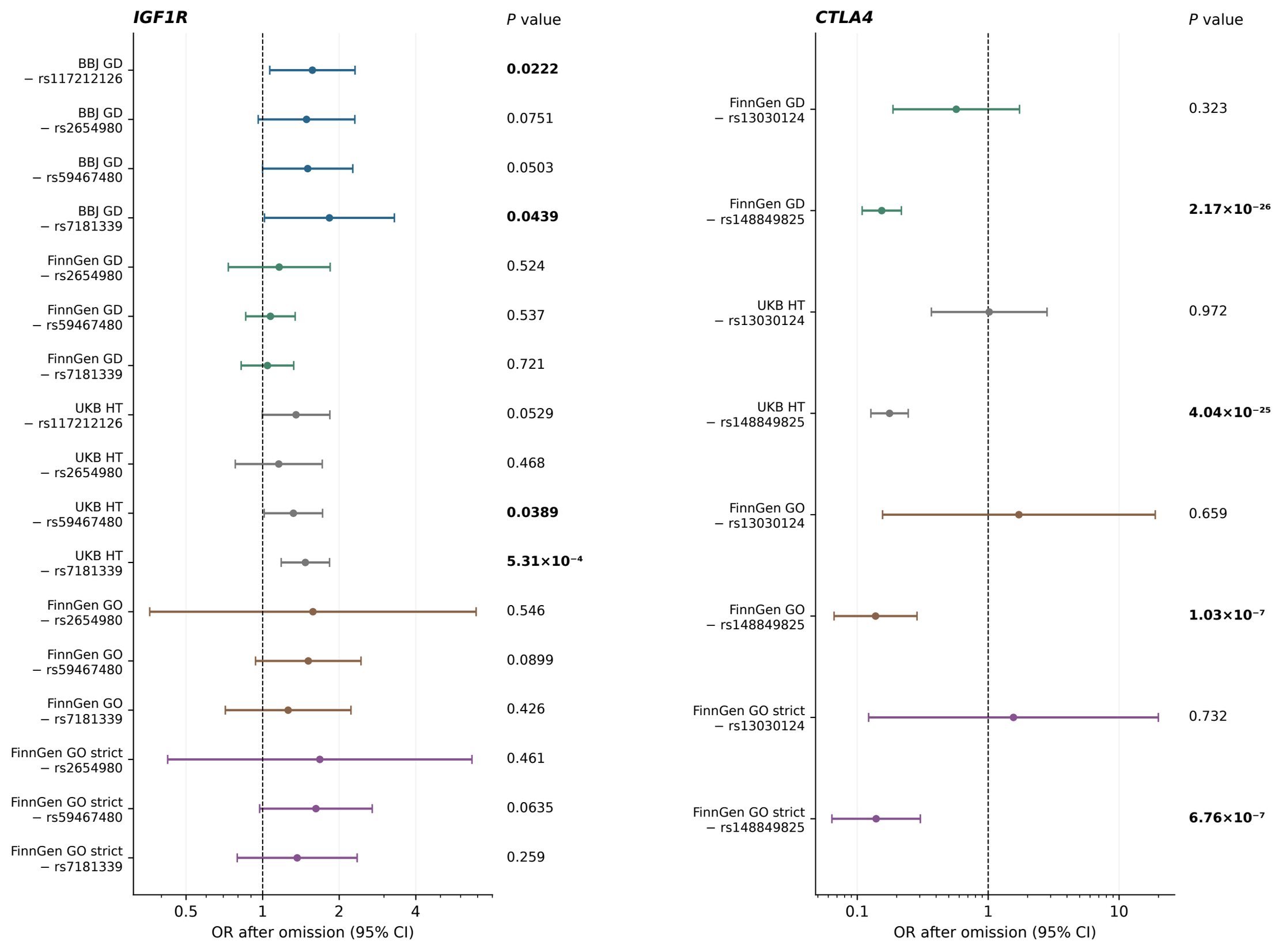
A Shared-variant probability (PP.H4)

	BBJ GD	FinnGen GD	UKB HT	FinnGen GO	FinnGen GO strict
TSHR	0.951	0.264	0.226	0.986	0.891
IGF1R	0.073	0.010	0.404	0.032	0.038
CTLA4	0.201	0.254	0.953	0.978	0.979
HLA-A	<0.001	<0.001	<0.001	<0.001	<0.001
HLA-DQA2	<0.001	<0.001	<0.001	<0.001	<0.001
C4A	<0.001	<0.001	<0.001	0.030	0.326
HSD3B7	0.636	0.011	0.095	0.026	0.046
TUBB	<0.001	<0.001	<0.001	<0.001	<0.001
VKORC1	0.388	0.011	0.030	0.030	0.048
TNFSF14	0.994	0.303	0.237	0.017	0.018
PRSS36	0.168	0.063	0.115	0.037	0.033
MAPKAPK5	<0.001	0.041	0.005	0.026	0.029
PSMB8	<0.001	<0.001	<0.001	<0.001	<0.001
IFNGR1	0.989	0.313	0.052	0.020	0.021
APOM	<0.001	<0.001	<0.001	0.006	0.227
NEU1	<0.001	<0.001	<0.001	0.003	0.061
IGLV2-18	0.029	0.020	0.024	0.082	0.100
IGLV3-12	0.019	0.022	0.024	0.083	0.101
TNF	<0.001	<0.001	<0.001	<0.001	<0.001
HIST1H4E	<0.001	<0.001	<0.001	0.004	0.007
LTA	<0.001	<0.001	<0.001	<0.001	<0.001
LTB	<0.001	<0.001	<0.001	<0.001	<0.001
C4B	<0.001	<0.001	<0.001	0.030	0.311
HLA-G	<0.001	<0.001	<0.001	<0.001	0.001

B *TSHR* prior sensitivity




TSHR
log₂ fold change = +2.59
Transcript abundance (TPM)
1.0
0.8
0.6
0.4
0.2
0.0
Control
(n = 1)
TED
(n = 4)
IGF1R
log₂ fold change = +0.80
Transcript abundance (TPM)
7
6
5
4
3
2
1
0
Control
(n = 1)
TED
(n = 4)
CTLA4
log₂ fold change = +1.58
Transcript abundance (TPM)
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Control
(n = 1)
TED
(n = 4)


IGF1R
P value
BBJ GD – rs117212126
0.0222
BBJ GD – rs2654980
0.0751
BBJ GD – rs59467480
0.0503
BBJ GD – rs7181339
0.0439
FinnGen GD – rs2654980
0.524
FinnGen GD – rs59467480
0.537
FinnGen GD – rs7181339
0.721
UKB HT – rs117212126
0.0529
UKB HT – rs2654980
0.468
UKB HT – rs59467480
0.0389
UKB HT – rs7181339
5.31×10⁻⁴
FinnGen GO – rs2654980
0.546
FinnGen GO – rs59467480
0.0899
FinnGen GO – rs7181339
0.426
FinnGen GO strict – rs2654980
0.461
FinnGen GO strict – rs59467480
0.0635
FinnGen GO strict – rs7181339
0.259
0.5
1
2
4
OR after omission (95% CI)
CTLA4
P value
FinnGen GD – rs13030124
0.323
FinnGen GD – rs148849825
2.17×10⁻²⁶
UKB HT – rs13030124
0.972
UKB HT – rs148849825
4.04×10⁻²⁵
FinnGen GO – rs13030124
0.659
FinnGen GO – rs148849825
1.03×10⁻⁷
FinnGen GO strict – rs13030124
0.732
FinnGen GO strict – rs148849825
6.76×10⁻⁷
0.1
1
10
OR after omission (95% CI)